

Super-platforms And Blockchain Business Models With IBM Gerry Cuomo

For this session, we have with us, Jerry Cuomo. IBM fellow and Vice President of Blockchain Technologies. With IBM since 1987, he had an extraordinary career. And in 2015 Jerry Cuomo became Vice President of Blockchain Technologies. He is also the co-author of a great book, which is called "Blockchain for Business."

With Jerry, we focus in this session about blockchain, enterprise blockchain, blockchain-based business models and in which ways might the blockchain really change how we do business. And as business people what playbook we need to develop to move toward that era.

What drove you to focus on a blockchain?

Jerry Cuomo: I've had the pleasure of working on transaction processing in my career at IBM, and I would say we are now with blockchain in the third generation of transaction process Generation One.

In fact, IBM was vibrantly involved in Generation One, which was the era of the mainframe transaction. And in fact, that era is still running as we speak. You know, with systems like Sabre running many of the airline and hotel reservations.

And then Generation Two, and this is where I got involved, with generation two which is web-based transaction processing. So we helped the likes of eBay create their web-based auctioning system.

Many people said that you wouldn't be able to really do a credit card transaction over the internet. But look, people laugh today, of course, you can. So you know, all banks and insurance companies and governments and everything in between have web-based, now mobile-based means to transact.

What got me involved in 2015 is, what was the next generation? Was there even going to be another generation in transaction processing, or would AI and IOT and Cloud make those go away.

And earlier, maybe in 2013, 2014 I saw Bitcoin, but something happened. In 2014 I saw Ethereum, and **really what impressed about Ethereum was smart contracts. And that's where I thought the value of blockchain could be applied to any industry.**

And you know, the very elegant thing about blockchain, it's about... It's as basic as the notion of teamwork. Meaning, a group working together is more likely to provide or produce a better outcome than any single member of that group working by themselves. It's pretty logical. And that's kind of at the center of blockchain.

It's this decentralization that brings trust to data.

So in 2015, again seeing Ethereum with smart contracts, we got this view that blockchain could be applied to any industry.

And that's set off the journey for what has now become IBM Blockchain, powered by a very compelling open-source project called Hyperledger and Hyperledger Fabric, which is, I'd say the blockchain technology that was made from the ground up for business.

So in 2015, it was a twinkle in my eye that this was possible. And I'm really glad here, a few years later, seeing the fruits of that starting to pay off.

What are the key elements of the Enterprise Blockchain?

Jerry Cuomo: Enterprise blockchain has many common characteristics of blockchain non-enterprise, I'm not sure what you would call the opposite of enterprise blockchain, but let's just say there are some base qualities to the blockchain, and there are some enrichments that make blockchain more enterprise savvy.

Blockchain in a nutshell

So the base characteristics of blockchain that apply to any type of blockchain, you will see a shared ledger that is not managed by a single administrator, it's managed by a group through a process of consensus.

And that when transactions are consented on, they are appended to a shared ledger in a way that is cryptographically secured with the prior block, **thereby making a chain that is resistant to tampering, and forms an audit log that becomes the center of trust.** And I think that is indigenous to blockchains.

A crash course in Enterprise Blockchain

But then enterprise blockchain and Hyperledger fabric was probably the first enterprise blockchain, not the only one today, but certainly the first. We set out, and we looked at four or five additional qualities.

- **The first is accountability.** So prior to this type of enterprise, blockchain members were anonymous in participation. And there was good rhyme and reason for that, for cryptocurrency emulating the property of cash.

Cash is a bearer instrument, no need necessarily to figure out who the person is, or who the institution is. But through proof of work, very clever, it was possible to gain trust in an institution that was participating in the blockchain without knowing who they are.

But in an enterprise to pass government regulations and things like in healthcare, HIPAA or, and for around privacy and identity, GDPR, the members must be accountable.

And that's where the notion of permission blockchain, so members are known to the network. That's number one.

- **Number two is while they're known, enterprises need to operate with confidentiality and privacy.** So looking at additional capabilities around blockchain for privacy and confidentiality, I mean, Hyperledger fabric supports a notion, if those users out there are familiar with Slack, and this notion of a channel where you can subdivide the ecosystem based on a particular topic. So Hyperledger fabric supports that. And that allows for more private... Not everyone in the network gets to broadcast the same transaction, but you can subdivide and add this level of confidentiality, privacy. That's number two.

- **Number three is needed for immense performance and scalability.** Enterprises have an insatiable appetite for performance. So performing and transacting at thousand transactions per second or more is a base requirement for the enterprise.
- **And the other one is finality, which means the network, once it arrives at an answer, the answer can't fork, it has to stay, it has to become final.** And with Ethereum and Bitcoin, there's been forks of the network. Certainly, we've worked very hard with Hyperledger fabric to ensure that a transaction once committed, is final.

- **And then last, the fifth is fault tolerance security.** So not every enterprise participating in a blockchain network is going to be the same. Meaning, some might have big IT budgets, some might have small IT budgets. Some may have great security architects, others lesser so. So the network has to keep running, even in the presence of actors that might be sloppy. So if you're not running the latest patches of an operating system, and you have a failure in your node, you can't take the network down. It just has to keep running. So accountability through permissions, privacy, confidentiality, performance, finality, and full tolerance security is what differentiates enterprise blockchain from all the other blockchains if that makes sense.

How do you need to change the mindset in a blockchain-driven business world?

Jerry Cuomo: Businesses have spent the last hundreds of years responsible for themselves. I talked about teamwork, and a group working together. **I like to say blockchain is a team sport.**

Businesses today aren't structured to work in teams. Yes, there's business to business, B2B. But even in a B2B transaction, you're still responsible. Like if you're in a B2B lending network, and you're depending on a third party to vet the authenticity of a particular client that you're lending money to, and they turn out to be a bad actor and the government comes after you for lending money to a bad actor, you can't say, "Oops, it wasn't me. It was someone I trusted in my network." No, you are responsible. You go to jail, they don't go to jail.

The big trick in the business playbook is to design business processes that can be worked across a team.

And that might sound difficult, but again, in a B2B world, we're already working together as a team.

I can mention shared reference data. This is an example from financial services where companies, financial services companies, usually hire intermediaries to distribute the results, the day end results of an exchange, to be able to record the reference data associated with the price of a stock, let's say.

It's not like my reference data is better than yours.

We all get the same reference data, but we all spend a lot of money in distributing reference data. So if we can just distribute it to ourselves, perhaps that's a more efficient way to do it, and we'd save money, right?

In a financial services situation, there's another example. Like if I get a fraudulent act to happen in my bank, I might want to be a good citizen and share that with some of the other banks so that the same bad actor doesn't cause fraud in their banks.

So setting up a network. Those are win-win networks. It involves teams sharing information, where they haven't necessarily shared before, or would hire intermediaries to disseminate that, adding friction.

- Change to the playbook number one.

But now with blockchain, you can create a business model that is shared across companies.

And again, initially cost savings. So I do think one of the biggest tricks is looking at those business models.

In 2019, I can go almost industry for the industry and talk about the top two or three that have emerged in that industry, where it's really not a debate anymore. Companies are collaborating on it.

- Change to the playbook number two is, I would say a variation of that, and that's governance.

So once you have a great idea that leverages the power of the group versus the individual company, how do you govern that? Like what are the rules, who are the referees, and what is the mechanism by which you manage and operate such a network? What are the obligations, what are the rewards?

Governance becomes the next big important factor in your playbook.

And I guess the other thing is, like any new technology, being able to dream big about what the technology could ultimately do to transform your industry. Think about some, of the quotes from Walmart in the food industry, working as part of the Food Trust network.

Again, not even the world's biggest food companies have complete visibility across the supply chain. It would take Walmart seven days in a test they did, they did to trace back a packaged mango from one of their stores to the farm in which that mango came from.

On this blockchain network, with more diverse organizations sharing data, they can now pinpoint, let's say a foodborne illness in 2.2 seconds. That's transforming.

The question is, is any transformative thought, what's the scope? Where do you start? Number one. So how big of a bite do you take out of the initial project? And some people might call that your MVP, what is your minimum viable product that's going to take the first bold step against that big idea?

The next big question in a blockchain, from a business playbook perspective, is who are you going to take that step with? You know, what is your MVE, what's your minimal viable ecosystem?

Doing it with one company is hardly an ecosystem, that's like an ecosystem of one. Although some companies are diverse, both geographically operating across the different countries and regulatory jurisdictions.

But you know, one company might be too small, 20 companies might be too big, it may take too long to get off the ground. Although a decentralized network does bring a better trust model, often it is the harder place to start.

So starting more centralized might be a better place to start. But the problem with centralized is you may start quicker, but then how do you convince others that if you're joining and you're number five to join, that members one, two, three and four don't have an unfair advantage, right? So you have to prove that out as you go forward.

So all that said, the playbook at some levels changes based on the plurality of blockchain.

You know, working with groups, but you don't have to take it to an extreme, where you're working with hundreds just to get your idea off the ground. You know, you can start with a group of three or four, just enough decentralization to get your idea moving such that everyone in your group has control, but no one is exclusively in control.

So I think that's the balance in the new blockchain-based business models that we see.

Gennaro Cufano: Thanks, Jerry, it was very interesting. Actually, as you were talking, I was thinking probably one of the most revolutionary things from the business standpoint is also the fact that we go behind the corporation because in a blockchain the whole purpose might be about creating an ecosystem of companies that might look like a platform business model. But in a platform business model, we still have a company. We think a case of Google, Amazon, Facebook. I mean, those companies still centralize most of the profits, still have control over the platform, still have the ones which are really managed interactions on those platforms. I mean, those are the ones who make the rules of the game. In a blockchain-based on a business model, instead, we go behind the intermediary. So it feels to me like we are making the platform not relevant anymore.

Does the blockchain make centralized platforms irrelevant? Or is it an evolution of them?

Jerry Cuomo: It's a different type of platform. I think it's still a platform. I mean, when you look at these networks, they look like platforms. The better ones have well-constructed APIs.

You know, people think about and say blockchain is web 3.0, it's the web of value versus the web of information.

And in order for web 3.0 to take root, it needs to at least be web 2.0. So the better networks look like platforms that are programmable, they have APIs, and in fact, **a really good platform based on a blockchain doesn't ever even say the word blockchain. It's just a valuable network.**

And if you talk about what, again, the obligations and rewards are for joining the network, and the fact that it was built with blockchain, and I think that is that an interesting conversation over dinner, but when you're sharing the value of it, it may or may not even have to come up.

But the economics are different. **The economics are different from today's more single company centered platform business models.** It's not to say that single companies for different types of networks aren't going to benefit, but I do think this really favors the ecosystem.

And I do think, again, we'll see, and are seeing, a new style of platform that looks from a programmability perspective like what we saw on web 2.0 platforms. But I think economics is more balanced around a consortium or a group versus an individual company.

What is a Minimum Viable Ecosystem (MVE) and how to kick it off?

Jerry Cuomo: Just to give you an example, the best ecosystems are the ones that are already there. And your best partners are the ones that you already have. I mean, IBM as a company, as a buyer of services, has about 20,000 suppliers.

And that's not unusual. I mean, I'm sure if you look at companies, there are probably companies that have 100,000 suppliers. But neighboring companies like us most likely have the same.

They're buyers for similar suppliers. And you know, our chief procurement officer knows many of the other chief procurement officers and companies that we do business with. So as a real story, our chief procurement officer is now working with, I'd say about a dozen other chief procurement officers and companies like IBM, to create a blockchain network to help the buyer-supplier relationship.

So that was a group of people starting initially with three companies, IBM and two others, with a business partner. And we had good relationships. I mean they were there, we had common pain that we felt, and inefficiencies of modern-day processes.

Once our chief procurement officer figured out that, and they asked us to prove that blockchain could address some of these inefficiencies, they were off to the races with three. But again, **I would always recommend starting less, more centralized than decentralized.**

I also think even when I've been in rooms with a dozen banks looking to join a consortium, and while that looks decentralized, the fact of the matter **in any group's surroundings, you're going to get the group leaders to emerge right away.**

And there'll be about two or three. So in the end, you typically start with two or three, whether it's just two or three, or two or three in a group of 12. It's just natural group behavior that we'll start that way.

So when you start with a smaller group, you just have to spend more time figuring out why let's say if you start with three, why would the fourth ever join? What's their obligation? What's their reward? If you take all the rewards as the first three, come on, you wouldn't join that club. So why would they join the club?

So you have to keep the door open, and that has to be factored in if you start more at centralized. If you start more decentralized, then you have to figure out how you ever get to your MVP.

If more than the three people that I mentioned who are usually vocal or vocal, if you have to answer to 12 lawyers, it's going to take you a very long time to get to your MVE completion.

The bigger MVEs usually are run by only a few people who have strong opinions. So, in the end, you're always typically working and starting a group with a small group, and building from there. So again, your results will vary based on the industry, and based on the actual use case. But that's my experience.

How does the blockchain inject trust in a business model?

Jerry Cuomo: You can say, "Jerry, why should I trust you?" And I can give you a list of things I've done in my life that make me trustworthy. And you can step back and consider those things to see whether that truly makes me a trustworthy person or not.

You know, again, you can look at my reputation and all of this kind of stuff, talk to other people, and you might get a good view. In a blockchain, it's mathematical, it's algorithmic. So you don't have to quote-unquote trust of someone doing that to their reputation.

You trust someone because it is very hard to tamper with the data. So you know, again, it stems from the actual technology, that trust emanates and grows from the technology. It's enforced through it. So very, very simple, a database has a single administrator.

I mean it wouldn't be right to compare blockchain completely to a database, but it's easier to reason this way. But you know, **a blockchain has multiple administrators, multiple businesses, each with a copy of the ledger, each getting to weigh in on the consensus process, each getting a copy of the ledger once they consent on a transaction.**

And you know, unlike a database that has commands like update and deletes, a blockchain is append-only, right. And once that block is appended, it's cryptographically linked to the previous block, creating a chain that is very hard to tamper with.

And that audit log becomes the foundation of trust. It's not something about necessarily about reputation, it's about that algorithm, and that ledger was when the data is in it, and we all consented, there it is. And we can sleep at night knowing that it was, that no single company went in and altered or doctored the data.

So the business trust is anchored in that algorithmic computational trust that is built. So it transcends from being hard to manage or hard to quantify reputational thought, into a highly quantifiable, algorithmic, provable thought.

So that's... And whether a business understands it or not is another story, but it does anchor that trust between businesses in a way that was very difficult to do before this technology.

How does economics change?

Jerry Cuomo: I think that that is a key aspect of the whole economical approach of blockchain. I actually think so much about blockchain is about a new economic. In fact, when you think about traditional Bitcoin, there's a very interesting economic there.

Like you give something in the form of energy, and you get a piece of a token back. And you know, I think around instance economics and also how tokenization plays in, I think is a very interesting concept. And we go over that in the book quite a bit. And I do think there is a role in enterprise blockchain for tokens, and for thinking about your business in that form.

I mean, digital assets are key to any kind of digital transformation. The problem is that for five digital applications today, their assets are represented in five completely different ways. **So with a token, you have a chance of reasoning about a digital asset across applications.**

I think something that Marley Gray from Microsoft is driving at the token taxonomy initiative, which is one of the enterprise Ethereum functions that IBM is participating in. I'm trying to help this economic effect really take root through standardization, so the TTI is an initiative to try to classify and create a taxonomy around tokens that could be applied to any technology, whether it's Hyperledger fabric or Ethereum or anything in between, to help kind of liberate the digital assets that are in these applications. And of course blockchain is ripe for this.

But even in blockchain, the ability right now, if you have 10 networks, you kind of have a tower of Babel effect going on. And there's no easy way for one network to reason with another network.

Thinking about the economics around this, and then thinking about how to quantify the economics in something. And I think tokenization comes to mind around that are key things to do. It's, again, a good blockchain idea, the best blockchain ideas start as a business idea. And from that really the fun begins.

And you know, I think blockchain doesn't change the art of business. It just, I would say, restructures it a bit around the group.

And the interesting thing is that we always heard about teamwork, and we all know teamwork provides the best results. I just think in business it's been hard to enforce that.

And I think with blockchain technology, it's the perfect peanut butter and chocolate, a perfect combination of good things coming together to form a new outcome.

Because I think businesses do work together in B2B networks, but you can't be responsible for others' actions. But in blockchain you can, the network can be a responsible entity. And I do think that expands an economic that that is pretty cool.

Key takeaways and quotes

- In a blockchain-based business, it's decentralization that brings trust to data.
- The key elements of an enterprise blockchain are accountability, confidentiality, and privacy, performance and scalability, finality, tolerance security.
- "The big trick in the business playbook is to design business processes that can be worked across a team."
- "Governance becomes the next big important factor in your playbook."
- "Economics is more balanced around a consortium or a group versus an individual company."
- "A blockchain has multiple administrators, multiple businesses, each with a copy of the ledger, each getting to weigh in on the consensus process, each getting a copy of the ledger once they consent on a transaction."
- "Blockchain doesn't change the art of business. It just, I would say, restructures it a bit around the group."
- "Businesses do work together in B2B networks, but you can't be responsible for others' actions. But in the blockchain you can, the network can be a responsible entity. And I do think that expands an economic that that is pretty cool."